

Creating specialized thresholds and classes for fire risk using machine learning methods

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Abstract

In this study, we examined forest-fire risk based on the localized climatological conditions of three distinct regions in Greece. Specifically, we observed that the widely used 30/30/30 rule for wildfire / forest-fire occurrence does not necessarily align with the climatic characteristics of different areas and has not been sufficiently validated through empirical research, leaving a notable gap in the wildfire /forest fire, protection literature. The motivation for this work stemmed from the occurrence of large wildfires during the non fire seasons in the United States, South Korea, Russia, and Japan, which highlighted the need for region specific hazard indicators / rules rather than universal thresholds. Thus, this study explores various machine learning techniques for assessing forest fire risk. The data used in this study have been selected from three different areas in Greece over the past 25 years (2000 - 2024). The first area of interest is Attica, the second area is Eastern Macedonia and Thrace, and the third area is Crete. For each of the aforementioned areas, a separate data file was created, an analysis was carried out, and specialized definitions - classes were created for the forest fire risk occurrence. To evaluate the region-specific hazard classes, this study compared a range of machine learning techniques, including classical methods (MLP, RBF, Random Forest) and methods based on Grammatical Evolution such as the FC2 feature construction algorithm and the GENCLASS rule induction system (publicly available tools employed for comparison purposes). Across all three datasets, GENCLASS demonstrated the most consistent overall performance among the tested techniques, closely followed by the feature construction method FC2 (MLP), while the remaining traditional approaches (MLP, RBF, and Random Forest) showed comparatively lower and more variable performance. However, this finding should be interpreted as exploratory due to the small effective number of regional datasets ($n = 3$ per model) and the non - significant Wilcoxon test results. Overall, the results support the relevance of the selected climatological attributes and region - specific definitions for fire-danger classification, suggesting that specific thresholds may provide a more suitable framework for wildfire risk assessment than the generic 30/30/30 rule.

Keywords: Fire risk; Machine learning; neural networks

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1. Introduction

This study presents a new rule for forest fire risk, which specifies and adapts the climatic and forest fire data with a new framework instead of the well known 30/30/30.

In short the rule 30/30/30 means that there is a risk of fire when the following conditions prevail: temperatures >30 , wind >30 , and humidity <30 . However, at this point we will briefly give a sample of analyses and experiments, with new definitions, which are specific to each region. For example, the dangerous mean temperature point for Attica is 23.6°C or 74.5°F , for Eastern Macedonia and Thrace is 20°C or 68.1°F , and for Crete is 20.17°C or 68.3°F . In other words, wildfires do not necessarily follow our predefined rules, however, our newly developed, region - with specific rules - adapted to the unique characteristics of each area, can contribute to more effective fire prevention and suppression efforts. In this context, it is important to highlight several real world case studies, that demonstrate the necessity of establishing new, region specific thresholds for wildfire ignition risk.

- A first example is the wildfire that broke out during the winter season, specifically in January 2025, in California, lasting 24 days [1]. Moreover, the study by Bouvier et al. (2025) indicates that an improved wildfire prevention model is required for periods traditionally considered as non hazardous for fire occurrence [2]. It is worth noting that, based on the meteorological data we retrieved from Visual Crossing for the region of California for the period 01/01/2025 - 31/01/2025, for this event, the mean temperature was approximately 5°C (42°F), with maximum temperatures reaching only 15°C (60°F) [3].
- A second example is the wildfire that broke out during the early spring season, specifically in March 2022, in South Korea [4]. Notably, by examining the temperature records for that specific period in the region of South Korea, the nighttime minimum temperatures dropped to -4°C (24.8°F), while daytime maximum temperatures ranged between 12°C (53.6°F) and 14°C (57.2°F) [5].
- A third example is the wildfire that occurred in May 2025 near the city of Chita in Russia [6]. During that period, temperatures were approximately 17.2°C (63°F), with the lowest recorded temperature reaching 4.4°C (40°F) [7].
- A fourth example is the winter wildfire that occurred in Ofunato, Japan, in February 2025 [8]. During that period, temperatures ranged from -1.6°C (29°F) to a maximum of 3.8°C (39°F) [9].

In summary, the aforementioned incidents represent only a subset of the wildfire events identified through the literature review, all of which fall outside the scope of the traditional 30/30/30 rule. This highlights the need to develop and examine new, region specific and context adapted metrics for assessing wildfire occurrence risk, based on meteorological variables and operational firefighting data.

In our search for studies related to our own work, we initially focused on identifying research that examined the application of the 30/30/30 rule and its reported outcomes, using the keywords: *30/30/30 rule fire risk*. The review of the retrieved studies indicated that only one verification oriented investigation of the rule has been conducted, and its findings were not satisfactory. Specifically, the study by Cuevas and Rodriguez (2021) highlights that the 30/30/30 rule does not account for the majority of large scale wildfires, explaining only about 10% of such events. In contrast, it appears more relevant for medium sized wildfires, accounting for approximately 66% of these cases. Moreover, temperature and relative humidity emerge as the most critical variables, as over 80% of the analyzed wildfires met these two conditions [10]. We will proceed by presenting studies and research works that examine the impact of meteorological variables and modeling approaches on forest fires, comparable to our own, and, where applicable, we will conclude by comparing their reported performance metrics with the findings of the present study.

The study by Cortez and Morais (2007) employed meteorological variables such as: temperature, relative humidity, rainfall, and wind, using an SVM model capable of predicting the burned area of small fires. Their approach achieved accurate predictions in

46% of the cases when an error of ≤ 1 ha was allowed, and in 61% of the cases when the admissible error increased to ≤ 2 ha [11]. The publication by Yunhao et al. (2007) focuses on comparing two wildfire risk indices: the Fire Potential Index (FPI) and the Integrated Fire Susceptibility Index (IFSI). Their experimental results, based on the Area Under the Curve (AUC) of the REC curves, showed that IFSI performed as the superior model with an accuracy of approximately 78%, whereas FPI achieved a prediction accuracy of about 73% [12].

Next on the research by Sakr et al. (2011) employed only two meteorological variables cumulative precipitation and relative humidity, together with two artificial intelligence techniques, artificial neural networks (ANN) and support vector machines (SVM), to accurately predict the risk of fire occurrence. Their results showed that, for the SVM model, each month included at least one iteration in which the testing accuracy exceeded 97% [13]. Analogous result were reported in the study by Agarwal et al. (2020) who applied the Gradient Boosting Tree algorithm using data from the National Fire Incident Reporting System (NFIRS) together with meteorological information from the National Oceanic and Atmospheric Administration (NOAA) to predict economic losses caused by fire incidents. Their results demonstrated strong predictive performance, achieving 93.3% accuracy on the full dataset and 97% accuracy on the test set [14]. The subsequent field study by Masinda et al. (2022) examined the daily assessment of wildfire danger in Northeastern China using the Canadian Forest Fire Weather Index (FWI). Their results indicated that the Canadian FWI performed well, with the Wilcoxon paired sample test showing satisfactory agreement in 72.7% of the cases [15]. The same modeling framework, combining the Canadian Forest Fire Weather Index (FWI) with the Initial Spread Index (ISI), was examined by Karali et al. (2023) in their assessment of hazardous wildfire prone conditions. Their findings indicated that lead time plays a decisive role in forecast performance, with the probability of correct prediction reaching approximately 60% [16]. In their study, Parvar et al. (2024) developed a wildfire risk map by integrating meteorological variables such as: Land Surface Temperature (LST), precipitation, and historical fire occurrence data. The performance of their model was statistically evaluated using the ROC curve, achieving an AUC value of 0.83%, indicating strong predictive capability [17]. **The literature review by Kudláčková et al. (2025) is also of particular interest, as it highlights—among other aspects—the need to incorporate specialized climatic variables into wildfire-risk assessment frameworks and to enhance early warning systems [18]. A similar approach was followed by the Australian Fire Danger Rating System, which had already adopted comparable principles in the 1950s and 1960s, when McArthur’s fire danger indices for forests and grasslands were developed, and later revised the system by introducing new thresholds [19]. Thus, a live trial of its Research Prototype (AFDRSRP), which is based on fire behavior thresholds, was conducted and evaluated between October 2017 and March 2018. Overall, the AFDRSRP outperformed the traditional Forest Fire Danger Index & Grassland Fire Danger Index (FFDI/GFDI) system, achieving 56% correct classifications compared to 43% [20]. Finally, India also attempted to develop a model tailored to its own conditions, using the Canadian fire prediction system as a foundation and incorporating human activity indicators, wildfire occurrence counts, and climatic variables into a neural-network-based framework [21].**

At this point, we will present, in a table 1, the results of the aforementioned studies in order to enable a direct comparison among them Table 1.

The remaining of this paper has the following structure: the Section 2 presents the used dataset and the new threshold values for wildfire occurrence, the Section 3 illustrates the experimental results finally the Section 4 It provides a concise presentation of the results, and discusses methodological limitations.

Table 1. Comparative overview of accuracy/performance results reported by previous studies on forest fire risk prediction, as identified in the literature review.

Authors names	Results
Diaz Cuevas, M. D. P., & Limones Rodriguez, N.[10]	10% accuracy (large fires) & 66% (medium fires)
Cortez, P., & Morais, A. D. J. R[11]	46% error (of ≤ 1ha) & 61% (≤ 2 ha)
Yunhao, C., Jing, L., & Guangxiong, P.[12]	IFSI accuracy (78%) & FPI (73%)
Sakr, G. E., Elhajj, I. H., & Mitri, G.[13]	SVM accuracy (97%)
Agarwal, P., Tang, J., Narayanan, A. N. L., & Zhuang, J[14]	Gradient Boosting Tree accuracy (93.3%) & test set (97%)
Masinda, M. M., Li, F., Qi, L., Sun, L., & Hu, T[15]	FWI Wilcoxon test accuracy (72.7%)
Karali, A., Varotsos, K. V., Giannakopoulos, C., Nastos, P. P., & Hatzaki, M[16].	FWI & ISI accuracy (60%)
Parvar, Z., Saeidi, S., & Mirkarimi, S.[17]	AUC value of 0.83

2. Materials and Methods

In this section, a detailed description of the datasets used as well as the machine - learning techniques applied in the experiments.

2.1. The dataset employed

For the purposes of this study, we combined two distinct data sources. On the one hand, to analyze the climatological conditions, we used Historical Weather Data from Visual Crossing [https:// www.visualcrossing.com/](https://www.visualcrossing.com/) [3] (last accessed on 02 February 2026). This platform compiles information from a range of authoritative sources, including national meteorological agencies, global climate datasets, and satellite observations. Access to the historical dataset required registration. We then used the data - access interface, to specify the region of interest, and the relevant time period. Data were subsequently retrieved through the free access tier, which allows up to 1,000 records per day. The procedure was generally straightforward, with the main limitation being the daily allowance of 1,000 records.

On the other hand, to analyze forest and wildfires, we relied on the open data provided by the Hellenic Fire Service [https:// www.fireservice.gr/el_GR/synola-dedomenon](https://www.fireservice.gr/el_GR/synola-dedomenon) (last accessed on 02 February 2026). The fire - related data are collected and updated annually by the individual fire stations responsible for reporting these incidents.

2.2. The dataset description

For this study, we obtained historical meteorological data covering the period from 2000 to 2024, providing a approximately 25 year record for analyzing fire duration in relation to climatological patterns. In addition, we selected regions located at different latitudes and longitudes across Greece, to provide a more comprehensive representation of the study area. The selected regions of interest were: Attica, Eastern Macedonia and Thrace, and Crete 1.

The climatological and fire dataset for all regions cover the period from 1 January 2000 to 31 December 2024. Specifically, Attica contains 9.131 climatological records and 2.736 fire records, Eastern Macedonia and Thrace contains 9.131 climatological records and 10.358 fire records, and Crete contains 9.121 climatological records and 6.036 fire record. The slight variation in the number of climatological records is attributable to missing values in the data retrieved from Visual Crossing, and did not affect the results of the analysis.



Figure 1. Map of Greece and Islands highlighting the three geographic study areas (Study Area 1: Attica, Study Area 2: Eastern Macedonia & Thrace, Study Area 3: Crete).

Regarding missing data in the Visual Crossing, the meteorological dataset for Attica contained 9.131 daily records (2000–2024), and the dataset for Eastern Macedonia & Thrace contained the same number of records for the same period. The dataset for Crete exhibited a small gap of 10 missing daily records during May 18–28, 2001. This gap does not affect the merged dataset or the class construction procedure, because during that specific period only two wildfire incidents occurred, both of which were short duration events and were not included in the analysis. Therefore, the missing data do not influence the threshold derivation, the class definitions, or the subsequent machine learning experiments.

Before proceeding with the merging of the two data files, the experiments focused exclusively on forest fire incidents in order to maintain homogeneity in our dataset with respect to fire behavior characteristics. In other words, we chose to proceed with the following categories : (9) Burned forests, (10) Burned forest area, and (11) Burned grove (small forest) Table 3. Following this refinement, Attica provided 1.859 forest fire records, Eastern Macedonia and Thrace 2.653, and Crete 2.789.

Afterwards, the climatological dataset merged with the forest fire dataset, retaining only the climate records corresponding to the fire ignition date. The merging process was performed combining the climatological data from Table 2 with the fire incident data from Table 3 based on the date field.

Table 2. Primary table of raw climatological variables (33 attributes) retrieved from the Visual Crossing platform for the period 2000–2024.

Number	Data Description	Values
1	Region	Area
2	Date	Year/month/day
3	Temp max (Fahrenheit)	°F
4	Temp min (Fahrenheit)	°F
5	Temp (Fahrenheit) daily mean average	°F
6	Feels like max (Fahrenheit)	°F
7	Feels like min (Fahrenheit)	°F
8	Feels like (Fahrenheit)	°F
9	Dew	°F
10	Humidity	%
11	Precipitation	in
12	Precip prob	%
13	Precip cover	%
14	Precip type	Type(s)
15	Snow	in
16	Snow depth	in
17	Windgust (miles)	mph
18	Windspeed (miles)	mph
19	Wind dir	(°)
20	Sea level	mb
21	Cloud cover	%
22	Visibility	mi
23	Solar radiation	(W/m ²)
24	Solar energy	(MJ/m ²)
25	Uvindex	(0-10)
26	Severisk	(0-100)
27	Sunrise	time
28	Sunset	time
29	Moon phase	(0-1)
30	Conditions	general observations
31	Descriptions	general observations day/night
32	Icon	summarized observations
33	Stations	Code for stations

Table 3. Primary table of raw operational forest fire incident data (16 attributes) as recorded by the Hellenic Fire Service.

Number	Data Description
1	Fire department
2	Forest department
3	Location
4	Region
5	Date of fire ignition
6	Time of fire ignition
7	Extinguish date
8	Extinguish time
9	Burned forests
10	Burned forests area
11	Burned grove (small forest)
12	Burned grassland
13	Burned swamps
14	Burned agricultural
15	Burned crops
16	Burned garbage dumps

Accordingly, we created three new datasets, one for each study region. Each region was examined and analyzed independently, resulting in a distinct fire risk classification for every area. Consequently, the variables processed in order to construct the class representing danger forest fire ignition, for all three datasets were the following: mean daily temperature, daily humidity values, precipitation occurrence, wind gusts, wind speed, wind direction, time of fire ignition, month of fire ignition, and fire duration, with the final output expressed as the Final_DangerClass_combination Table 4. In the end, in order to ensure impartiality in our experiments, we removed the Fire Duration feature [35], since the model would otherwise be able to see the answer, as the accuracy for all datasets reached 100%.

Table 4. Final derived variables/classes obtained after processing the raw data, used to construct the target variable Final_DangerClass_combination.

Number	Data
1	Temp Special Class (daily mean average)
2	Humidity Special Class
3	Precipitation Basic Class
4	Windgust Special Class
5	Windspeed Special Class
6	Wind dir Special Class
7	Time Class
8	Month class
9	Fire duration (Removed)
10	Final_DangerClass_combination

2.2.1. Construction of region-specific fire-danger thresholds and classes.

To identify the thresholds of the climatological features associated with long-duration wildfire / forest-fire events, the following procedure was applied independently to each region and its corresponding dataset.

The first step involved defining the critical duration threshold for long-duration forest fires. In the Greek context, this threshold is set at 360 minutes[35], since if a wildfire is not brought under control within the first few hours, it tends to spread rapidly beyond containment and may continue burning for several days [36]. Accordingly, wildfires

with a duration of up to 360 minutes, were assigned to (class 0), whereas wildfires with a duration of 361 minutes or more were assigned assigned to the long duration, (class 1).

Subsequently, a systematic data-driven procedure, based on statistical mean estimation and iterative refinement, was applied to determine the hazardous values for each feature, as in Table 4. Specifically, the mean value of each climatological variable was first computed. Each initial cutoff was then evaluated by filtering the data according to the fire-duration attribute, in order to assess its ability to distinguish long-duration (class 1) from short-duration fires (class 0).

For example, in the Attica dataset, the mean temperature observed during wildfire events was 74.80°F. Accordingly, an initial classification rule was applied whereby temperatures up to 74.80°F were assigned to class 0, while temperatures of 74.81°F and above were assigned to class 1. This preliminary cutoff was then evaluated against the fire-duration feature to assess the extent to which class 1 values corresponded to long-duration fires. In this case, inconsistencies were identified, and the threshold refinement procedure was continued. Through this process, the final temperature threshold for Attica was determined to be 73.7°F.

An analogous refinement procedure was applied to all remaining climatological variables and across all regional datasets, yielding the final set of hazardous and non hazardous thresholds presented in Tables 5, 6, and 7.

In summary, this study followed a data-driven to identify hazardous climatological values rather than relying on a traditional climatological analysis. Meteorological data and operational fire-incident records were merged for the same time period. Long-duration wildfires were then filtered, and the relevant variables temperature, humidity, precipitation probability, wind speed, wind gusts, wind direction, ignition time, and month, were isolated. New empirical hazard thresholds associated with long-duration wildfire events were subsequently computed.

Based on these values, binary classes (0/1) were defined for each feature, where 1 denotes a hazardous value and 0 a non- hazardous value. Subsequently, a new dataset was constructed containing the transformed feature columns and the corresponding final danger class, and a ten - fold cross validation procedure was applied to evaluate the experimental results.

Thus, a regionally adapted forest-fire danger index was developed based on real climatological and operational data. Unlike the uniform 30/30/30 rule, the proposed index uses thresholds and weights tailored to the specific characteristics of each region. Since, the final danger class is derived from the transformed input variables, the classification experiments are interpreted as assessing the reproducibility of the proposed rule, with independent operational validation remains necessary.

2.2.2. *Impacts and consequences of forest-fire disasters.*

We next provide a concise overview of the consequences and broader impacts associated with wildfires / forest-fires. To begin with, wildfires substantially elevate air pollution levels. According to the World Health Organization, wildfire smoke constitutes a major source of toxic gaseous and particulate pollutants, as well as an important contributor to greenhouse gas emissions [22]. In the Figure 2, the carbon dioxide emissions (in tons) released into the atmosphere from wildfires in Greece during the period 2003–2025 are shown, as estimated by the Global Wildfire Information System.

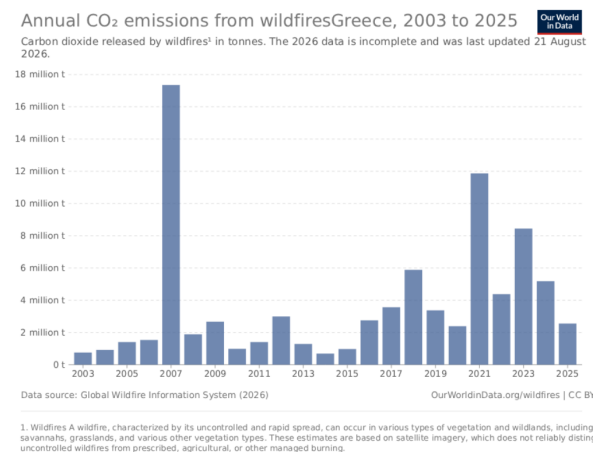


Figure 2. Chart of carbon emissions from wildfires in Greece during the period 2003–2025.

In addition to air pollution, wildfires also affect a country’s economic stability, as illustrated in the Figure 3. The chart illustrates economic losses from disasters as a percentage of gross domestic product (GDP), with wildfire related damages shown in orange.

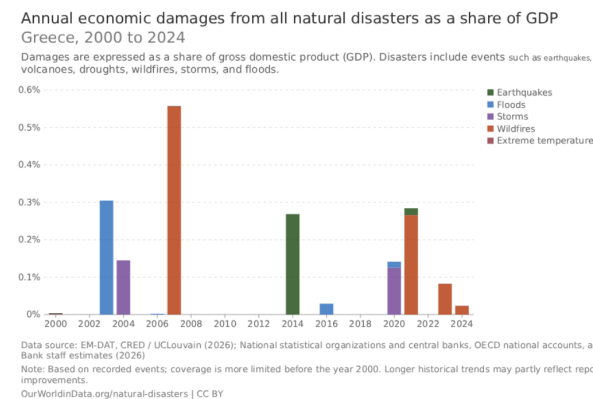


Figure 3. Chart of economic damages from all natural disasters during the period 2000 - 2025

In parallel, the number of people affected by a wildfire often reaches into the thousands, including individuals sustaining physical injuries, those requiring immediate assistance, and people left homeless who need temporary shelter because their homes were destroyed or severely damaged during the disaster, Figure 4.

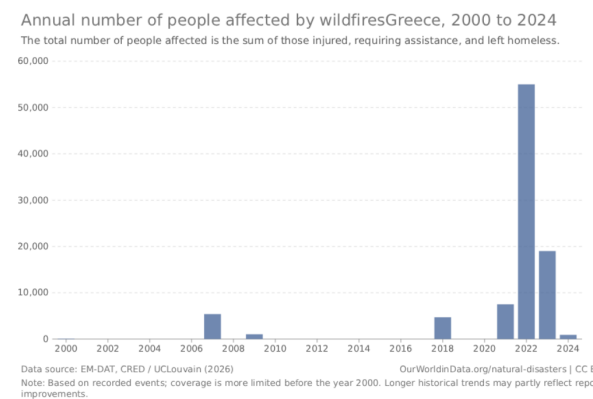


Figure 4. Chart of people affected from forest fires during the period 2000 - 2024

In closing, when discussing the impacts of wildfires, it is essential to acknowledge the loss of human life, the most severe and irreversible of all consequences, Figure 5. In summary, reducing the societal and environmental impacts of wildfires brings to the forefront the need for precise, region specific definitions of climatological thresholds that favor the development of long-duration wildfire events.

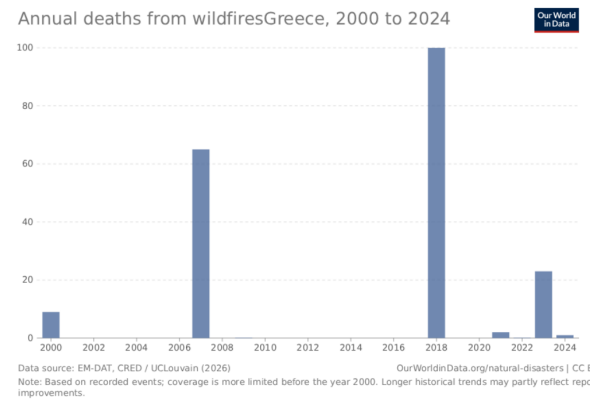


Figure 5. Chart of annual deaths from forest fires during the period 2000 - 2024

Next, we describe the data used for each region separately.

2.2.3. Attica area and data description

The regional unit of Attica covers an area of 3,808.1 km², with Athens as its capital. Between 2001 and 2025, Attica experienced a loss of 22 kha (220 km²) of tree cover attributable to wildfires, along with an additional 3.3 kha (33 km²) lost due to other disturbance factors. The most severe year within this period was 2021, during which fires accounted for 8.5 kha (85 km²) of tree cover loss, representing 99% of the total loss recorded that year, according to the global forest watch [23].

A brief description of the Attica dataset follows. The final dataset consisted of 1.859 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination. The classes followed a binary scheme, where 0 denoted a non hazardous value, and 1 denoted as a hazardous value, in the following Table 5.

Table 5. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Attica region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
Temp Special Class (daily mean average)	< (73.6°F) (lower)	> 73.7°F (or 23.6°C) (higher)
Humidity Special Class	> 49.4% (higher)	< 49.3% (lower)
Precipitation Basic Class	rain	no rain
Windgust Special Class	< 10miles (lower)	> 10.01miles (higher)
Windspeed Special Class	< 16miles (lower)	> 16.1miles (higher)
Wind dir Special Class	(S, SE, E, W)	(SW, N, NE, NW)
Time Class	< 12:30 (before)	> 12:31 (after)
Month class	(1,2,3,4,5,9,10,11,12)	(6,7,8*)
Final_DangerClass_combination	844 data	1.015 data

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Attica, as follows: temperature $\geq 73.7^{\circ}\text{F}$, humidity $\leq 49.3\%$, 0% precipitation, wind gust speed ≥ 10.01 mph, wind speed ≥ 16.1 mph, wind direction in the sectors SW, N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August*.

2.2.4. Eastern Macedonia and Thrace area and data description

The regional unit of Eastern Macedonia and Thrace, covers an area of 14.158 km², with Komotini as its capital. Between 2001 to 2025, Eastern Macedonia and Thrace experienced a loss of 40 kha (400 km²) of tree cover attributable to wildfires, along with an additional 11 kha (110 km²) lost due to other disturbance factors. The most severe year within this period was 2023, during which fires accounted for 31 kha (310 km²) of tree cover loss, representing 94% of the total loss recorded that year, according to the global forest watch[23].

A brief description of the Eastern Macedonia and Thrace dataset follows. The final dataset consisted of 2.653 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination. The classes follow the same binary scheme as previously. A concise summary table of these variables is presented in Table 6.

Table 6. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Eastern Macedonia & Thrace region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
Temp Special Class (daily mean average)	< 68°F (lower)	>68.1°F (or 20°C) (higher)
Humidity Special Class	> 64.1% (higher)	< 64% (lower)
Precipitation Basic Class	rain	no rain
Windgust Special Class	< 7miles (lower)	> 7.1miles (higher)
Windspeed Special Class	< 6miles (lower)	> 6.1miles (higher)
Wind dir Special Class	(W,NW,N)	(E,SE,SW,S,NE)
Time Class	< = 11:30	11:31 =>
Month class	(1,2,3,4,9,11,12)	(5,6,7,8,10*)
Final_DangerClass_combination	1.093	1.560

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Eastern Macedonia and Thrace, as follows: temperature $\geq 68.1^\circ\text{F}$, humidity $\leq 64.1\%$, 0% precipitation, wind gust speed ≥ 7.1 mph, wind speed ≥ 6.1 mph, wind direction in the sectors E, S, SW, SE, and NE, ignition time from 11:31 onwards, and hazardous months corresponding to May, June, July, August and October.

2.2.5. Crete area and data description

The regional unit of Crete covers an area of 8.450 km², with Herakleio as its capital. Between 2001 to 2025 Crete experienced a loss of 1.3 kha (13 km²) of tree cover attributable to wildfires, along with an additional 3.2 kha (32 km²) lost due to other disturbance factors. The most severe year within this period was 2008 during which fires accounted for 210 ha (2.1 km²) of tree cover loss, representing 44%, of the total loss recorded that year, according to the global forest watch [23].

A brief description of the Crete datasets follows in Table 7.

Table 7. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Crete region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
Temp Special Class (daily mean average)	< 70°F (lower)	>70.1°F (or 20.17°C) (higher)
Humidity Special Class	> 66.1% (higher)	< 66% (lower)
Precipitation Basic Class	rain	no rain
Windgust Special Class	< 9miles (lower)	> 9.1miles (higher)
Windspeed Special Class	< 16miles (lower)	> 16.1miles (higher)
Wind dir Special Class	(S, SW, SE, E, W)	(N, NE, NW)
Time Class	< = 12:30	12:31 =>
Month class	(1,2,3,4,5,9,10,11,12)	(6,7,8*)
Final_DangerClass_combination	1.384	1.405

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Crete, as follows: temperature ≥ 70.1 °F, humidity $\leq 66\%$, 0% precipitation, wind gust speed ≥ 9.1 mph, wind speed ≥ 16.1 mph, wind direction in the sectors N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August .

2.2.6. The feature construction method

For a subset of the conducted experiments, a feature construction technique [43] involving two artificial features was employed. A brief description of the applied procedure is provided below. This method generates artificial features from the original input variables through the Grammatical Evolution process. The resulting features represent non-linear transformations of the initial features. In recent years, this method has been applied to several real-world problems, including earthquakes [24], flash flood [25], fetal heart rate [27], spam recognition [28] etc. The extended version of BNF grammar used during the feature construction process is outlined in Figure 6

```

S ::= <expr>      (0)
<expr> ::= (<expr> <op> <expr>) (0)
          | <func> (<expr>) (1)
          | <terminal> (2)
<op> ::= + (0)
        | - (1)
        | * (2)
        | / (3)
<func> ::= sin (0)
        | cos (1)
        | exp (2)
        | log (3)
<terminal> ::= <xlist> (0)
            | <digitlist>.<digitlist> (1)
<xlist> ::= x1 (0)
          | x2 (1)
          | .....
          | xn (n-1)
<digitlist> ::= <digit> (0)
              | <digit><digit> (1)
              | <digit><digit><digit> (2)
<digit> ::= 0 (0)
          | 1 (1)
          | .....
          | 9 (9)

```

Figure 6. The extended BNF grammar used in the feature construction process

The main steps of the used algorithm are as follows:

1. Initialization step.

- (a) **Define** the number of used chromosomes N_c .
- (b) **Define** the maximum number of allowed generations N_g .
- (c) **Set** the selection rate $p_s \in [0, 1]$ and the mutation rate $p_m \in [0, 1]$.
- (d) **Set** as N_f the number of desired features that will be created.
- (e) **Set** $k = 0$, the generation number.

2. Fitness calculation step.

3. For $i = 1, \dots, N_c$ do

(a) **Create** with the assistance of Grammatical Evolution, a set of N_f artificial features from the original ones, for chromosome g_i .

(b) **Transform** the original train set using the previously produced features. Represent the new set as $TR = (x_{g_i,j}, t_j), j = 1, \dots, M$

(c) **Apply** a machine learning model denoted as C on set TR and train this model and denote as $C(x)$ the output of this model for any input pattern x .

(d) **Calculate** the fitness f_i as:

$$f_i = \sum_{j=1}^M (C(x_{g_i,j}) - t_j)^2 \quad (1)$$

The Radial Basis Function (RBF) networks [29,30] were used as the machine learning models $C(x)$ in the current work. This machine learning model was chosen because of the significantly shorter training time it requires compared to other machine learning models.

4. End For

5. Genetic operations step. Perform the same genetic operators as in the case of construction neural networks, discussed previously.

6. Termination check step.

- (a) **Set** $k = k + 1$
- (b) **If** $k \leq N_g$ then go to Fitness Calculation Step.

7. Application to the test set.

- (a) **Obtain** the chromosome g^* with the lowest fitness value.
- (b) **Create** the N_f artificial features that correspond to this chromosome
- (c) **Apply** the N_f features to the train set and produce the mapped training set $TR = (x_{g^*,j}, t_j), j = 1, \dots, M$
- (d) **Train** a machine learning model on the produced training set. An artificial neural network [31,32] with $H = 10$ processing nodes is used in the current work. This neural network was trained using a BFGS variant of Powell [39].

- (e) **Apply** the new features to the test set of the objective problem and create the set $TT = (x_{g_{ij}}, t_j), j = 1, \dots, K$
- (f) **Apply** the machine learning model on set TT and report the test error.

3. Results

For the purposes of our experiments, we employed the available optimization package Optimus[26], which can be downloaded from <https://github.com/itsoulos/GlobalOptimus> (accessed on 20 June 2026). For the feature construction process used in the FC2 (RF) and FC2 (MLP) algorithms, we utilized the freely available programming tool QFc[33], accessible at <https://github.com/itsoulos/QFc> (accessed on 20 June 2026). In addition, the GENCLASS [34] system was employed in our experiments. GENCLASS is a free software framework entirely written in ANSI C++, available by <https://github.com/itsoulos/GenClass/> (accessed on 20 June 2026) designed to construct classification programs in a C like programming language for the purpose of classifying input data. It operates as an evolutionary rule induction method, generating interpretable decision rules through genetic search mechanisms.

Each experiment was executed 30 times, using different seeds for the random number generator in every run. **For each experiment, a complete ten - fold cross - validation procedure was performed, resulting in 30×10 train-test evaluations for each algorithm and dataset.** The proposed feature construction method was applied independently to each fold, and the average classification errors on the corresponding test sets were computed. For the experimental tables the following notation was used:

1. The column DATASET denotes the dataset corresponding to a regions where the machine learning methods were applied. Attica contains 1.859 records, of which 844 correspond to class 0 and 1.015 to class 1. Eastern Macedonia & Thrace contains 2.653 records, of which 1.093 correspond to class 0 and 1.560 to class 1. Crete contains 2.789 records, of which 1.384 correspond to class 0 and 1.409 to class 1.
2. The column MLP (BFGS) represents the application of an artificial neural network [37,38] with 10 processing nodes trained with the BFGS optimization method [39].
3. The column RBF stands for the usage of a Radial Basis Function (RBF) network [40,41] with 10 processing nodes.
4. The column RF denotes the application of the Random Forest model [42].
5. The column FC2 (RF) represents the construction of two artificial features using the Grammatical Evolution feature construction technique [43]. The Radial Basis Function (RBF) machine learning model was employed for feature construction, while the generated features were evaluated using Random Forests.
6. The column FC2 (MLP) stands for the construction of two artificial features using the same feature construction technique. The evaluation of the constructed features was performed using a neural network trained with the BFGS method.
7. The column GENCLASS represents the incorporation of the GENCLASS method to produce classification rules.
8. The row AVERAGE represents the average classification error for each dataset and method.

3.1. Classification error

The results demonstrated in Table 8 that the model GENCLASS achieved the lower error accuracy in predicting the final danger class, indicating that the newly developed fire

danger rule is fully identifiable and predictable based on the combined climatological and operational fire incident data.

Table 8. Experimental results using the machine learning methods on various regions. The numbers in cells represent average classification error.

DATASET	MLP (BFGS)	RBF	RF	FC2 (RF)	FC2 (MLP)	GENCLASS
Attica	13.67%	14.42%	13.93%	13.01%	12.82%	12.43%
Eastern Macedonia & Thrace	12.51%	15.34%	12.21%	11.93%	11.87%	11.47%
Crete	16.39%	17.82%	15.96%	16.15%	15.56%	14.86%
AVERAGE	14.19%	15.86%	14.03%	13.70%	13.42%	12.92%

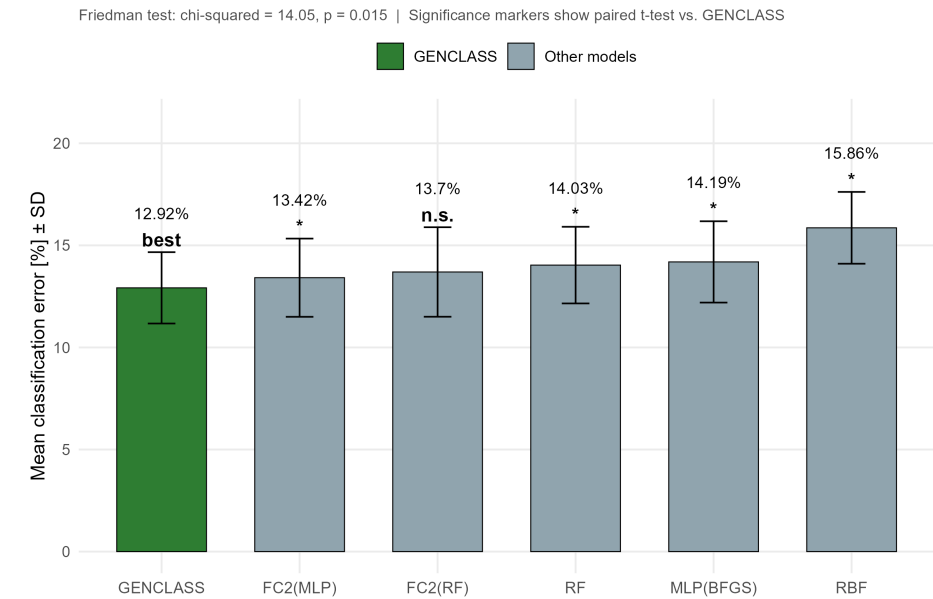


Figure 7. Average classification error by , across the three study regions

Figure 7 presents, in bar-chart form, the mean classification error rate (%) achieved by each of the six evaluated machine learning models, averaged across the three study regions (Attica, Eastern Macedonia & Thrace, and Crete), with error bars representing one standard deviation. GENCLASS is visually distinguished from the remaining five models and serves as the reference model against which pairwise comparisons are drawn. It attained the lowest mean classification error (12.92%), followed in ascending order by FC2 (MLP) (13.42%), FC2 (RF) (13.70%), RF (14.03%), MLP (BFGS) (14.19%), and RBF (15.86%), the latter exhibiting both the highest mean error and the greatest variability across regions. The figure further reports the result of a Friedman test ($\chi^2 = 14.05$, $df = 5$, $p = 0.015$), a non-parametric procedure appropriate for comparing multiple related samples, which indicates that the observed differences in performance among the six models are unlikely to have arisen by chance. Significance markers displayed above each bar correspond to paired t-test comparisons against GENCLASS: all models differ significantly from GENCLASS ($p < 0.05$) with the exception of FC2 (RF), which is marked as not significant (n.s.), suggesting that its performance is statistically comparable to that of GENCLASS despite its marginally higher mean error.

Table 9. Statistical comparison of classification error, average ranking, and significance tests (Wilcoxon and paired t-test) for the six machine learning models relative to GENCLASS as the reference model.

Model	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs GENCLASS	Wilcoxon p	Paired p	Significance
GENCLASS	12.92	1.75	11.47	14.86	1	0			Reference
FC2(MLP)	13.42	1.92	11.87	15.56	2	-0.5	0.1814	0.0395	*
FC2(RF)	13.7	2.19	11.93	16.15	3.33	-0.78	0.1814	0.0955	n.s.
RF	14.03	1.88	12.21	15.96	4	-1.11	0.1814	0.0367	*
MLP(BFGS)	14.19	1.99	12.51	16.39	4.67	-1.27	0.1814	0.0123	*
RBF	15.86	1.76	14.42	17.82	6	-2.94	0.1814	0.0324	*

Friedman test: $\chi^2 = 14.048$, $df = 5$, $p\text{-value} = 0.0153$

Nemenyi critical difference ($\alpha = 0.05$): 4.353

Significance vs. GENCLASS (paired t-test): *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant

Note: $n = 3$ regions per technique; results are indicative given the small sample size.

Table 9 complements Figure 7 by providing a detailed statistical summary of model performance across the three regions ($n = 3$ per model). For each model, the table reports the mean, standard deviation, minimum, and maximum classification error, together with the average rank obtained across regions, the mean difference relative to GENCLASS, and the corresponding significance estimates derived from both the Wilcoxon signed-rank test and the paired t-test. GENCLASS consistently ranks first (average rank = 1) with the lowest mean error and a comparatively narrow standard deviation ($SD = 1.75$), whereas RBF ranks last (average rank = 6) with both the highest mean error and the widest range of observed values. The Wilcoxon test yields an identical p-value (0.1814) for all comparisons, which is not statistically significant; this outcome reflects the limited statistical power of the test given the very small sample size ($n = 3$ paired observations per comparison). In contrast, the paired t-test, being a parametric test with greater sensitivity under these conditions, identifies statistically significant differences ($p < 0.05$) between GENCLASS and all other models except FC2 (RF) ($p = 0.0955$, n.s.), which again emerges as the closest competitor to GENCLASS. The table footnote reports the overall Friedman test statistic ($\chi^2 = 14.048$, $df = 5$, $p = 0.0153$) and the Nemenyi critical difference (4.353) used for post-hoc multiple comparisons, while explicitly cautioning that, owing to the small sample size (three regions per technique), the results should be regarded as indicative rather than conclusive.

Taken together, Figure 7 and Table 9 converge on the conclusion that GENCLASS systematically outperforms the remaining five models, with the superiority being statistically robust, as confirmed by the paired t-test, against every model except FC2 (RF), which stands out as its most competitive alternative.

3.2. Precision and recall values

Precision and recall are evaluation metrics used to assess the performance of a machine learning model. Precision expresses the proportion of predicted positive cases that are actually correct, while recall indicates the proportion of true positive cases that the model successfully identifies. Together, these metrics provide essential insight into the effectiveness of a model in classification tasks. Precision is calculated as:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

Precision reflects the model's ability to avoid false positives, making it a key indicator of prediction reliability. In other words, the higher the value, the fewer the errors.

The Recall value confirms the model's ability to correctly identify the positive class, reflecting how many of the actual positive instances were successfully detected. As with

precision and recall, the higher the value, the more successful the classification. Recall is calculated as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{3}$$

The following table presents the average values of the precision and recall measures for each dataset and for every method Table 10. In this comparison as well, we observe the consistent performance of all methods. The best values are achieved by RF, FC2 (RF), FC2 (MLP), and GENCLASS, while for the Eastern Macedonia & Thrace dataset, MLP (BFGS) stands out.

Table 10. Comparative summary of average precision and recall values (%) for each machine learning technique, by study region (Attica, Eastern Macedonia & Thrace, Crete).

		MLP (BFGS)	RBF	RF	FC2 (RF)	FC2 (MLP)	GENCLASS
Attica	Precision	87.98%	85.43%	88.07%	88.24%	88%	88.50%
	Recall	87.14%	85.32%	87.33%	87.32%	88.02%	88.10%
Eastern Macedonia	Precision	89.04%	83.51%	89.45%	87.92%	88.08%	89%
	Recall	88.05%	83.82%	88.37%	89.08%	89.08%	88.9%
Crete	Precision	84.46%	82.09%	85.07%	84.95%	84.46%	85.07%
	Recall	84.39%	82.09%	85.03%	84.88%	84.52%	85.50%

The precision and recall values presented in Table 10 confirm the pattern already observed for classification error, showing consistently high and stable performance across all three regions. For the Attica dataset, both metrics ranged between 85.32% and 88.50%, with GENCLASS, FC2 (RF), and RF achieving the strongest results, while RBF again recorded the lowest performance, at approximately 85%. The Eastern Macedonia & Thrace dataset exhibited the highest overall performance levels among the three regions, ranging from 83.51% to 89.45%, with RF, MLP (BFGS), and GENCLASS standing out in precision, and FC2 (RF) and FC2 (MLP) achieving the highest recall values. In contrast, the Crete dataset showed comparatively lower performance levels, ranging from 82.09% to 85.50%, although the relative ranking of the techniques remained consistent with the other two regions, with GENCLASS and RF standing out. Across all three datasets, RBF consistently underperformed relative to the other five techniques, whereas GENCLASS, RF, and the feature construction methods FC2 (RF) and FC2 (MLP) maintained the most stable and competitive precision and recall values, corroborating the reliability of the region specific fire danger thresholds developed in this study.

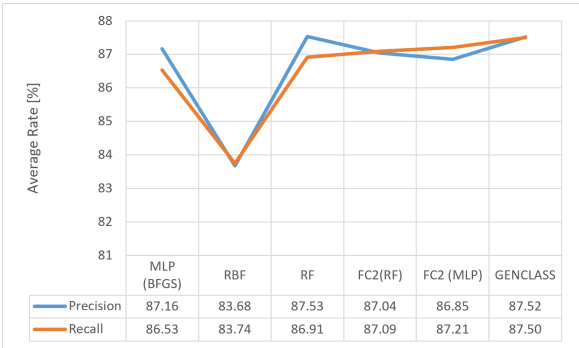


Figure 8. Line chart of the average precision and recall (%) across the three study regions, for each machine learning model.

Figure 8 summarizes these findings, showing that GENCLASS achieves the highest average recall across the three regions (87.50%) and an average precision (87.52%) essentially

equivalent to that of RF, the top-performing model on this metric (87.53%), while RBF remains the weakest performing technique throughout.

4. Conclusions

This study developed and validated region specific fire danger classification schemes for three climatically distinct areas of Greece, namely Attica, Eastern Macedonia & Thrace, and Crete, using twenty five years (2000-2024) of combined climatological and operational fire incident data. Rather than relying on the generic 30/30/30 rule, specialized threshold values for temperature, humidity, wind gust, wind speed, wind direction, ignition time, and month were derived independently for each region, reflecting their distinct climatic profiles. **Therefore, each region was treated independently: the climatological and operational data of Attica, Eastern Macedonia & Thrace, and Crete were processed separately, and the threshold construction (hazard class) procedure was applied within each regional dataset. As a result, no cross - region information leakage occurs.**

Six machine learning techniques were evaluated across all three datasets using ten fold cross-validation: two classical approaches (MLP trained with BFGS, and RBF networks), Random Forest (RF), two Grammatical Evolution based feature construction methods (FC2 (RF) and FC2 (MLP)), and GENCLASS, an evolutionary rule induction method also grounded in Grammatical Evolution. The results consistently demonstrated a clear performance hierarchy: GENCLASS achieved the lowest average classification error (12.92%) and the highest average precision and recall values across all three regions, confirming that the fire danger rule constructed in this study is **encouraging and emerges directly** from climatological and operational data. FC2 (MLP) and FC2 (RF) followed closely, ranking second and third respectively in average classification error (13.42% and 13.70%), while the classical techniques (RF, MLP (BFGS), and particularly RBF) exhibited comparatively higher and more variable error rates, with RBF consistently ranking last across all three regions in both classification error and precision/recall metrics.

This pattern indicates that techniques grounded in Grammatical Evolution, whether through automated feature construction (FC2) or direct rule induction (GENCLASS), are better suited to capturing the complex, region specific relationships between meteorological conditions and wildfire occurrence than conventional neural network or ensemble based approaches. In particular, the **strong performance** of GENCLASS suggests that the interpretable decision rules it generates are not only competitive in predictive accuracy but also offer a transparent and operationally meaningful representation of fire danger, a property that black box models such as MLP and RBF cannot provide. These findings validate the climatological attributes and the region specific definitions developed for the classification of fire danger, confirming that tailored, evolutionarily derived thresholds provide a more reliable and robust framework for wildfire risk assessment than generic, one size fits all rules.

However, it is crucial to acknowledge that these findings are limited to the three regions examined in this study (Attica, Eastern Macedonia & Thrace, and Crete). The generalizability of the region-specific classification schemes and the demonstrated superiority of Grammatical Evolution-based methods cannot be assumed for other Greek regions without external validation, particularly given the relatively small effective sample size on which these conclusions are based. Thus, while the results are encouraging for the examined areas, they represent a promising but not yet fully generalizable framework that requires validation across additional regions before broader operational deployment.

Building on these findings, several directions merit further investigation. First, the application of the proposed methodology to additional Greek regions with different cli-

matic and topographic characteristics would help establish whether the observed **strong performance** of Grammatical Evolution based techniques generalizes beyond the three areas examined here. Second, incorporating additional predictive features, such as vegetation type, fuel moisture content, terrain slope, and proximity to human activity, could further improve the discriminative power of the constructed classes, particularly for the Crete dataset, which exhibited comparatively lower performance across all techniques. Third, future work could explore the integration of the fire duration variable, excluded in this study to preserve model impartiality, as a secondary prediction target once ignition risk has been established, thereby extending the framework from a binary danger classifier to a more comprehensive operational decision support tool. Finally, given the consistently strong performance of GENCLASS, further research should focus on refining and extending the interpretability of the generated rule sets, potentially enabling their direct adoption by fire prevention services as a practical, region specific alternative to the traditional 30/30/30 rule.

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